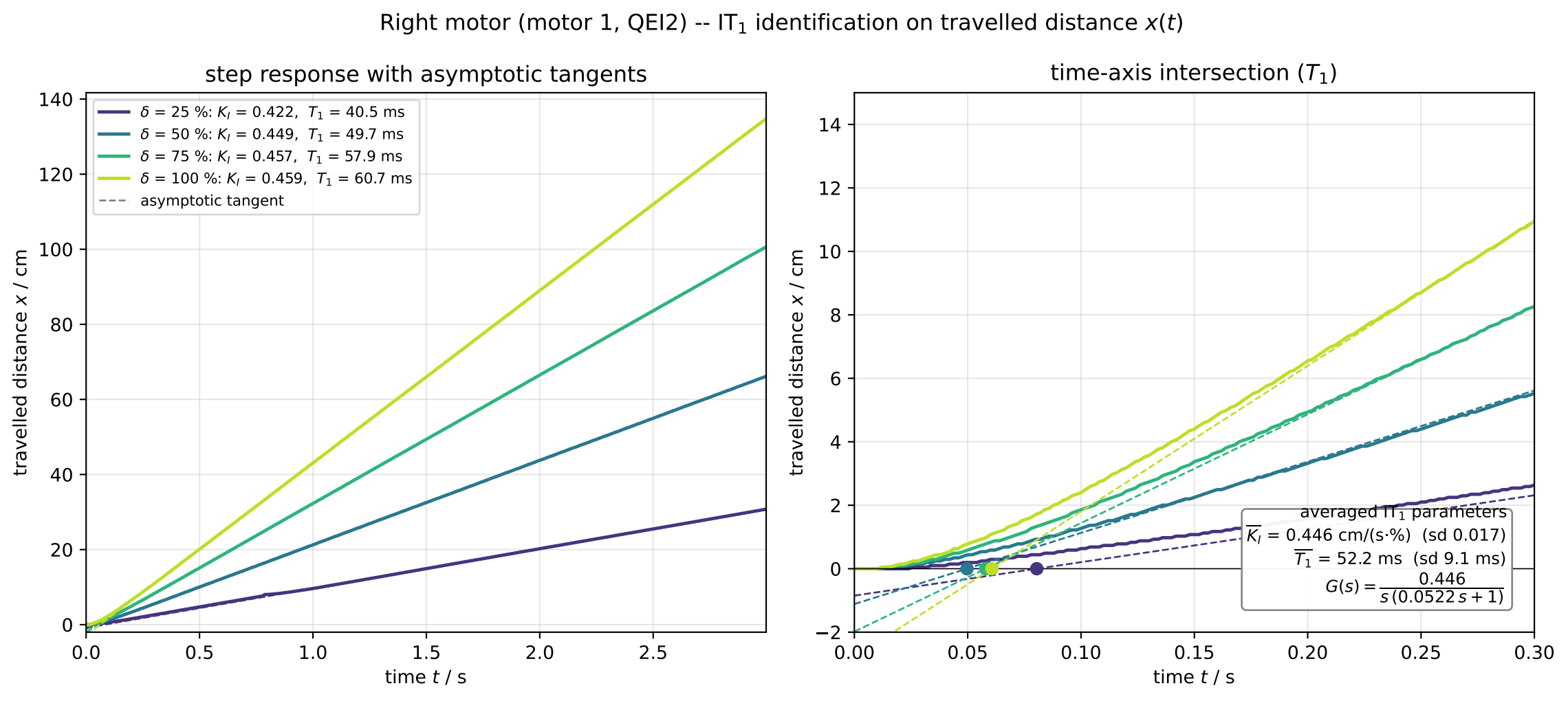
Right motor (motor 1, QEI2) -- IT$_1$ identification on travelled distance $x(t)$
step response with asymptotic tangents
δ = 25 %: K_I = 0.422, T_1 = 40.5 ms
δ = 50 %: K_I = 0.449, T_1 = 49.7 ms
δ = 75 %: K_I = 0.457, T_1 = 57.9 ms
δ = 100 %: K_I = 0.459, T_1 = 60.7 ms
asymptotic tangent
travelled distance x / cm
time t / s
time-axis intersection (T_1)
travelled distance x / cm
time t / s
averaged IT$_1$ parameters
$\overline{K_I}$ = 0.446 cm/(s·%) (sd 0.017)
$\overline{T_1}$ = 52.2 ms (sd 9.1 ms)
$G(s) = \frac{0.446}{s(0.0522\,s+1)}$